

## **SIMATS ENGINEERING ASSESSMENT TOOL 2**

**COURSE NAME: CSA04 COURSE CODE: Operating Systems**

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### **COURSE OUTCOME COVERED**

***CO1: Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)***

***Assessment Tool Weightage: 35%***

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**QUESTIONS: 5**

**Total Marks:25**

Annexure A

### **Concept Mapping**

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#### ***Code of Conduct:***

*I M. Santosh(192521007) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

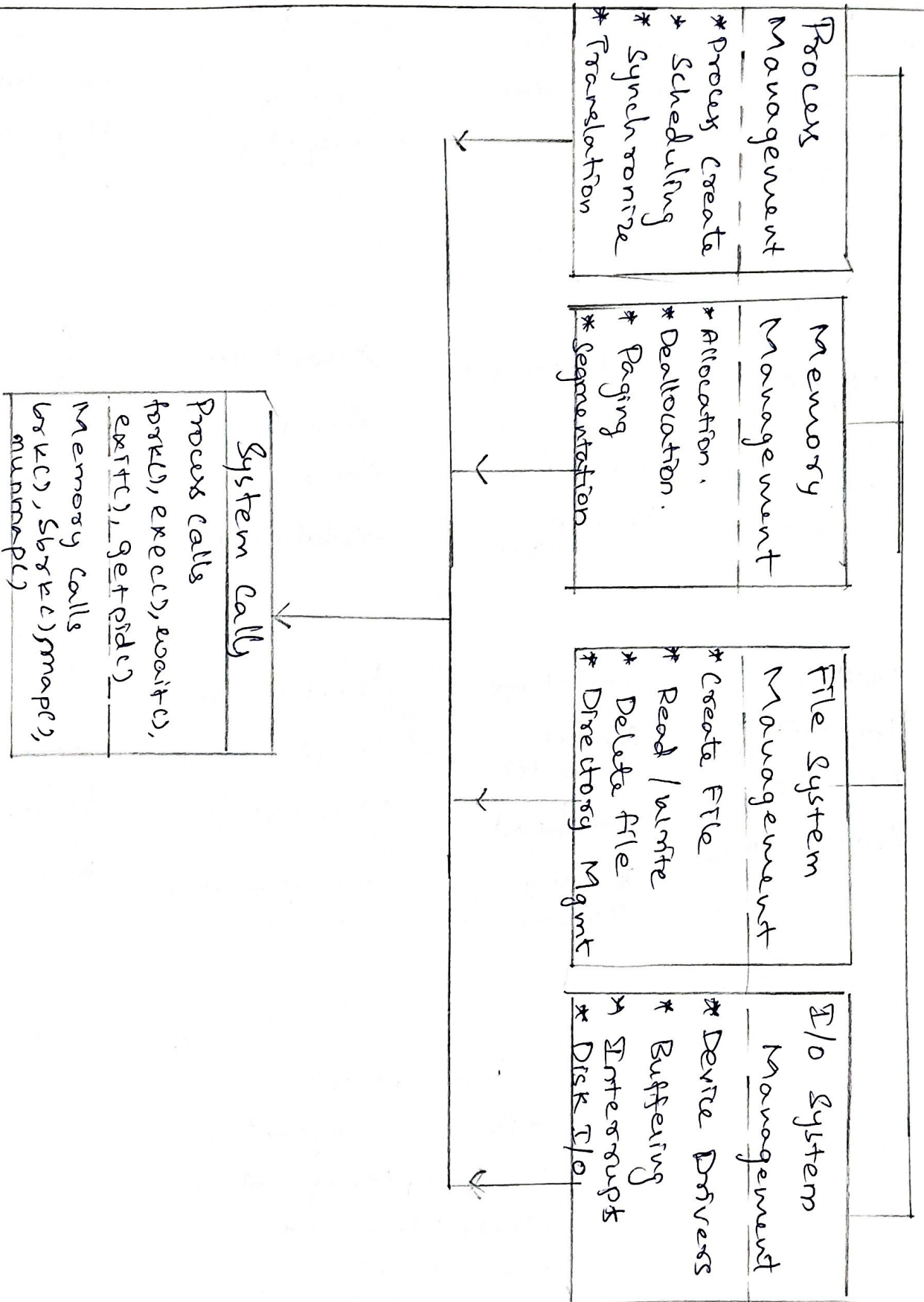
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

M. Santosh  
192521007

CO1  
AT-2  
Concept Mapping  
Operating Systems  
CSAO401

Operating Systems



2)

# Operating Systems

**Batch OS**  
Processes Jobs  
in Batches  
No Interaction  
High  
Throughput

**Advantages**  
High CPU  
Utilization  
Simple

**Limitations**  
Long Waiting  
No User  
Interaction

**Time-Sharing OS**  
Shares CPU  
among Users  
Interactive

**Advantages**  
Multi-user  
Quick Response  
Fair CPU  
share

**Limitations**  
Context  
Switch  
Overhead  
Security  
Issues

**Distributed OS**  
Multiple  
Computers  
Resource  
sharing  
Network  
based

**Advantages**  
Resource  
Sharing  
Scalability

**Limitations**  
Complex Design  
Network failure  
Synchronization

**Real-Time OS**  
Immediate  
Response  
Deadline-based  
High Reliability

**Advantages**  
Predictable  
Fast Execution  
Reliable

**Limitations**  
Expensive  
Limited Use  
Resource heavy

**Similarities**

- \* Process Management
- \* Memory Management
- \* Device Management
- \* CPU Scheduling
- \* Resource Allocation